Aufgabe 3: DHCP

Um DHCP-Pakete einzufangen, habe ich meine WLAN-Verbindung erstmal deaktiviert und dann wieder gestartet. Dadurch konnte ich diese Pakete in Wireshark sehen.

No.	Time	Source	Destination	Protocol	Length	Info
248	59.945899	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xe642e643
366	62.873317	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xe642e643
510	64.747127	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xe642e643
746	68.697401	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xe642e643
851	73.824788	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0x1e2be03c
891	76.816224	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0x1e2be03c
892	76.832756	192.168.0.1	192.168.0.217	DHCP	358	DHCP ACK - Transaction ID 0x1e2be03c
917	81.661061	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0x48efa751
988	84.644847	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0x48efa751
989	84.670350	192.168.0.1	192.168.0.217	DHCP	358	DHCP ACK - Transaction ID 0x48efa751

```
Frame 891: 364 bytes on wire (2912 bits), 364 bytes captured (2912 bits) on interface \Device\NPF_{F8245424-00FE-4520-87F4-B9F953BDFE6E}, id 0
Ethernet II, Src: Intel_a0:00:b4 (00:28:f8:a0:00:b4), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
User Datagram Protocol, Src Port: 68, Dst Port: 67
Dynamic Host Configuration Protocol (Request)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x1e2be03c
    Seconds elapsed: 3
    Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: Intel_a0:00:b4 (00:28:f8:a0:00:b4)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
    Option: (53) DHCP Message Type (Request)
    Option: (61) Client identifier
    Option: (50) Requested IP Address (192.168.0.217)
    Option: (12) Host Name
    Option: (81) Client Fully Qualified Domain Name
    Option: (60) Vendor class identifier
    Option: (55) Parameter Request List
    Option: (255) End
```

dhcp

No.	Time	Source	Destination	Protocol	Length	Info
248	59.945899	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xe642e643
366	62.873317	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xe642e643
510	64.747127	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xe642e643
746	68.697401	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xe642e643
851	73.824788	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0x1e2be03c
891	76.816224	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0x1e2be03c
892	76.832756	192.168.0.1	192.168.0.217	DHCP	358	DHCP ACK - Transaction ID 0x1e2be03c
917	81.661061	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0x48efa751
988	84.644847	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0x48efa751
989	84.670350	192.168.0.1	192.168.0.217	DHCP	358	DHCP ACK - Transaction ID 0x48efa751

```
Frame 892: 358 bytes on wire (2864 bits), 358 bytes captured (2864 bits) on interface \Device\NPF_{F8245424-00FE-4520-87F4-B9F953BDFE6E}, id 0
Ethernet II, Src: MS-NLB-PhysServer-16_18:b1:1c:44 (02:10:18:b1:1c:44), Dst: Intel_a0:00:b4 (00:28:f8:a0:00:b4)
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.217
User Datagram Protocol, Src Port: 67, Dst Port: 68
Dynamic Host Configuration Protocol (ACK)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x1e2be03c
    Seconds elapsed: 3
    Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 192.168.0.217
    Next server IP address: 192.168.0.1
    Relay agent IP address: 0.0.0.0
    Client MAC address: Intel_a0:00:b4 (00:28:f8:a0:00:b4)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
    Option: (53) DHCP Message Type (ACK)
    Option: (54) DHCP Server Identifier (192.168.0.1)
    Option: (51) IP Address Lease Time
    Option: (58) Renewal Time Value
    Option: (59) Rebinding Time Value
    Option: (1) Subnet Mask (255.255.255.0)
    Option: (28) Broadcast Address (192.168.0.255)
    Option: (3) Router
    Option: (81) Client Fully Qualified Domain Name
```

DHCP verwendet UDP mit den Ports 67 (Server) und 68 (Client). Es wird mit Broadcasts gearbeitet, da der Client zu Beginn keine IP-Adresse kennt.

Ein vollständiger DHCP-Vorgang besteht normalerweise aus vier Phasen (DORA): Discover, Offer, Request, ACK. Ich konnte nicht alle vier Pakete einfangen, nur Request und ACK.

Die DHCP-Optionen enthalten wichtige Netzwerkinformationen, wie Client IP Adresse, Subnetzmaske, Gateway, DNS-Server, Lease-Dauer usw. Jeder DHCP-Vorgang wird durch eine Transaction ID identifiziert.